9L Implement BWMatching

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Implement BWMatching
Count all occurrences of a collection of patterns in a text.

Input: A string BWT(*Text*), followed by a collection of strings *Patterns*.
Output: A list of integers, where the *i*-th integer corresponds to the number of substring matches of the *i*-th member of *Patterns* in *Text*.

```
                $BANANAS
                ANANAS$B
   2  1         ANAS$BAN
                AS$BANAN
BANANAS$        BANANAS$
 ANA  AS        NANAS$BA
  ANA           NAS$BANA
                S$BANANA
```

Formatting

Input: A string BWT(*Text*) and a space-separated list of strings *Patterns*.
Output: A space-separated list of integers, where the *i*-th integer corresponds to the number of substring matches of the *i*-th member of *Patterns* in *Text*.

Constraints

- The length of BWT(*Text*) will be between 1 and 10^4.
- The number of strings in *Patterns* will be between 1 and 10^1.
- The length of any given string in *Patterns* will be between 1 and 10^4.

Test Cases

Case 1

Description: The sample dataset is not actually run on your code.

Input:

```
TCCTCTATGAGATCCTATTCTATGAAACCTTCA$GACCAAAATTCTCCGGC
CCT CAC GAG CAG ATC
```

Output:

```
2 1 1 0 1
```

Case 2

Description: A larger dataset of the same size as that provided by the randomized autograder.